BT5110 Project: Tour de France Database and Analytics System

AY26/27 Semester 1

Background

The **Tour de France** is an annual professional road cycling competition held primarily in France, with occasional routes passing through neighbouring countries. The race spans approximately three weeks in July and covers more 3,500 kilometres. It consists of a sequence of stages, each of which may be either an individual time trial or a mass-start race. The overall winner of the Tour de France is the rider who completes all stages in the shortest cumulative time.

In addition to the **General Classification (GC)**, which determines the overall winner, the Tour de France features several other classifications, including:

- **Points Classification** (for sprinters)
- **King of the Mountains Classification** (for climbers)
- **Young Rider Classification** (for eligible younger riders)

The competition is contested by professional cycling teams. Teams typically consist of riders who work together throughout the competition to support each other's efforts. The Tour de France includes a mix of WorldTour teams and wildcard invitations. The teams come from various countries, although many are based in Europe. Some of the prominent cycling nations, including France, Italy, Belgium, Spain, and the Netherlands, often have multiple teams in the Tour de France.

A Tour de France cycling team normally fields a squad of **eight riders** with specialised roles such as climbers, sprinters, domestiques (support riders). The riders in a team may come from various countries that are different from the country represented by the team. Each team has a designated leader who is the primary contender for the General Classification. Team members work together strategically to support their designated leader in pursuit of individual and team objectives.

Application Specification

1. **Country**
   A country is uniquely identified by the **ISO 3166-1 alpha-3 code** [1]. This is a 3-letter country code. For each country, we record the country code and country name. The database must allow countries to exist even if they have no registered team and no registered rider.

2. **Team**
   A team is uniquely identified by its **team name**. For each team, we record the team name and associated country. Each team belongs to exactly one country. A team must have at least one rider. Teams without riders should not be recorded in the database.

3. **Rider**

   A rider is uniquely identified by their **bib number**. For each rider, we record the bib number, rider name, date of birth, team and country. Each rider belongs to exactly one team and represents exactly one country. A rider's country does not have to be the same as the team's country. For example, a French team may include a German rider.

4. **Race and Stages**

   The Tour de France race consists of multiple stages. In each stage, teams race from one location to another. A location is identified by its **location name**. For each location, we record the location name and country. Each location belongs to exactly one country. Only locations that are used as the start or finish of at least one stage should be stored.

   Each stage takes place on a single day. It starts at one location and finishes at another location. There are different types of stages depending on the types of the environment. Stage types include **flat, hilly**, **mountain**, and **individual time-trial**. For each stage, we record the stage id, stage date, start location, end location, distance covered, and stage type.

5. **Stage Results**

   A result is recorded for every rider who completes a stage. We record the finish time taken in seconds. Since multiple riders may finish at the same time, we also record the rank to capture the order in which riders cross the finish line. Additionally, time bonuses are awarded at the end of each stage for the first few riders to cross the finish line. Potentially, penalties may also be incurred during a stage. The adjusted time is calculated as:

   Adjusted time = Finish time – Time bonus + Time penalty

6. **Rider Exits**

   A rider may withdraw from the race before the start of a particular stage. We record the reason for the rider's exit. If a rider exits at some stage S, no stage results should exist for that rider from stage S onwards.

7. **Awards**

   The General Classification winner is the rider with the **lowest cumulative adjusted time** across all completed stages is the winner. The Best Team award is given to the team with the lowest aggregate time, calculated using the three fastest riders from the team. You may explore other types of awards from the website [2].

Project Tasks

Task 1. Design an ER Diagram (10%)

Construct an Entity-Relationship (ER) diagram for this application. Your ER diagram should include the following entities: Country, Team, Rider, Location, Stage, Result, Exit, Classification/Award. The diagram should clearly show attributes including primary keys, relationships, cardinalities, participation constraints, weak or associative entities (if applicable). The design should capture the requirements and business rules described in the specification. Examples include countries may exist even if they have no team or no rider, teams must have at least one rider, riders belong to exactly one team and one country, rider nationality may differ from team nationality, stages are tied to start and finish locations, rider exits are associated with both riders and stages.

Due: Monday 21 September 2026

Task 2. Translate ER Diagram to Relational Schema (10%)

Convert the ER diagram into a relational database schema. For each table, specify the attributes, primary key, foreign keys, and integrity constraints. Apply normalization up to **Third Normal Form (3NF)** or **Boyce-Codd Normal Form (BCNF).** Justify key design decisions such as separation of entities into different tables, handling of many-to-many relationships, elimination of redundancy and update anomalies.

Due: Monday 12 October 2026

Task 3. Develop and Demo an OLAP Dashboard (15%)

Develop an analytical dashboard using a suitable business intelligence tool (e.g., Tableau, Power BI, or a web-based BI platform). The dashboard should support OLAP-style analysis and include the following features:

a. Drill-Down Analysis. Enable users to explore results at different levels of detail, such as by stage**,** rider**,** or team.

b. Classification Comparison. Compare rankings across different classification including General Classification, Points Classification**,** King of the Mountains Classification**,** Young Rider Classification.

c. Aggregate Performance Analysis. Analyse cumulative race times by rider, team and classification.

d. Country Participation Analysis. Investigate number of teams per country, number of riders per country.

e. Dashboard Design and Presentation. Design a clear, intuitive, and visually appealing dashboard. Use an organised layout, appropriate use of colours, visualisations and interactive features (where applicable) to communicate key insights.

Students are encouraged to go beyond the above requirements by performing additional analyses, incorporating relevant visualisations, and presenting meaningful insights derived from the data.

Due: Wednesday 11 November 2026

Task 4. Final Report (10%)

Your final report should contain the following sections:

1. ER Diagram. Output from Task 1.
2. Relational Schema and Normalization. Output from Task 2, including normalization justification.
3. Dashboard Design and OLAP Analysis. Output from Task 3, including screenshots and explanations of supported analysis.
4. Star Schema vs. Normalized Schema:
    - Implement a star schema consisting of fact tables (e.g. fact_results) and dimension tables (e.g. dim_rider, dim_team, dim_stage, dim_country).
    - Run representative analytical queries (e.g., cumulative time per rider, best team classification, stage-by-stage performance analysis).
    - Compare query complexity and response time.
    - Discuss trade-offs between highly normalized schema and star schema.

Due: Friday 13 November 2026

References

[1] ISO 3166-1 alpha-3. https://en.wikipedia.org/wiki/ISO_3166-1_alpha-3. [Online; last accessed 2025].

[2] Tour de France 2025. https://franceletour.com/. [Online; last accessed 2025].